

Integration Technique: The Complex Exponential Substitution

The Method

When faced with integrals of the form $\int e^{f(x)} \cos(g(x)) dx$ or $\int e^{f(x)} \sin(g(x)) dx$, where the integrand is resistant to standard integration by parts, we can transition to the complex plane. We construct a complex integral J such that our desired integral I is either its real or imaginary part, utilizing Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$.

The Direct Extraction Shortcut

When taking the real or imaginary part at the end of the calculation, **do not** expand the complex exponentials into their sine and cosine components too early. Instead, algebraic manipulation of the exponents $e^{iA} \cdot e^{iB} = e^{i(A+B)}$ inherently handles all trigonometric angle sum and difference identities for you. If you have a final complex expression of the form $C \cdot e^{i(\text{angle})}$, you can directly read the components:

$$\begin{aligned}\operatorname{Re}\left(e^{i(\text{angle})}\right) &= \cos(\text{angle}) \\ \operatorname{Im}\left(e^{i(\text{angle})}\right) &= \sin(\text{angle})\end{aligned}$$

If your expression is multiplied by $-i$, remember that $-i(x + iy) = y - ix$, so the *real* part of the total expression will be the *imaginary* part of the complex exponential.

Problem Set and Solutions

Problem 1: The Base Product

Evaluate:

$$I = \int e^{\cos x} \cos(2x + \sin x) dx$$

Solution:

Construct the complex integral J such that $I = \operatorname{Re}(J)$.

$$\begin{aligned}J &= \int e^{\cos x} e^{i(2x + \sin x)} dx \\ &= \int e^{\cos x + i \sin x} e^{2ix} dx \\ &= \int e^{e^{ix}} e^{2ix} dx\end{aligned}$$

Let $u = e^{ix}$. Then $du = ie^{ix} dx \implies e^{ix} dx = -i du$.

$$\begin{aligned} J &= \int e^u \cdot u \cdot (-i du) \\ &= -i \int u e^u du \end{aligned}$$

Using integration by parts:

$$J = -ie^u(u - 1)$$

Substitute $u = e^{ix}$ back into the expression:

$$\begin{aligned} J &= -ie^{e^{ix}}(e^{ix} - 1) \\ &= -ie^{\cos x + i \sin x}(e^{ix} - 1) \\ &= -ie^{\cos x} (e^{i(x+\sin x)} - e^{i \sin x}) \end{aligned}$$

We require the real part of J . Since the entire bracket is multiplied by $-i$, the real part of J will be $e^{\cos x}$ times the *imaginary* part of the bracketed expression. Using our shortcut:

$$\operatorname{Im}(e^{i(x+\sin x)}) = \sin(x + \sin x) \quad \text{and} \quad \operatorname{Im}(e^{i \sin x}) = \sin(\sin x)$$

Therefore, directly extracting the result:

$$\mathbf{I} = e^{\cos \mathbf{x}} [\sin(\mathbf{x} + \sin \mathbf{x}) - \sin(\sin \mathbf{x})] + \mathbf{C}$$

Problem 2: The Polynomial-Exponential Blend

Evaluate:

$$I = \int e^{\cos x} [\cos(2x + \sin x) + 2 \cos(x + \sin x)] dx$$

Solution:

Construct the complex integral J such that $I = \operatorname{Re}(J)$.

$$\begin{aligned} J &= \int e^{\cos x} [e^{i(2x+\sin x)} + 2e^{i(x+\sin x)}] dx \\ &= \int e^{\cos x + i \sin x} (e^{2ix} + 2e^{ix}) dx \\ &= \int e^{e^{ix}} (e^{2ix} + 2e^{ix}) dx \end{aligned}$$

Let $u = e^{ix}$, so $du = ie^{ix} dx$, giving $dx = \frac{du}{iu}$. Notice $e^{2ix} = u^2$ and $e^{ix} = u$:

$$\begin{aligned} J &= \int e^u (u^2 + 2u) \frac{du}{iu} \\ &= -i \int e^u (u + 2) du \end{aligned}$$

Applying integration by parts:

$$J = -ie^u(u + 1)$$

Substitute $u = e^{ix}$ back in:

$$\begin{aligned} J &= -ie^{e^{ix}}(e^{ix} + 1) \\ &= -ie^{\cos x} \left(e^{i(x+\sin x)} + e^{i \sin x} \right) \end{aligned}$$

Again, we need the real part of J , which is $e^{\cos x}$ times the imaginary part of the bracketed expression.

$$\mathbf{I} = \mathbf{e}^{\cos \mathbf{x}} [\sin(\mathbf{x} + \sin \mathbf{x}) + \sin(\sin \mathbf{x})] + \mathbf{C}$$

Problem 3: The Rational-Exponential Twist

Evaluate:

$$I = \int e^{\cos x} [(1 - \cos x) \cos(\sin x) - \sin x \sin(\sin x)] dx$$

Solution:

We assemble the complex integral J such that $I = \operatorname{Re}(J)$. Note that $(1 - \cos x)$ is the real part and $\sin x$ is the imaginary part of a complex expression.

$$\begin{aligned} J &= \int e^{\cos x} [(1 - \cos x)(\cos(\sin x) + i \sin(\sin x)) + i \sin x(\cos(\sin x) + i \sin(\sin x))] dx \\ &= \int e^{\cos x} e^{i \sin x} (1 - \cos x + i \sin x) dx \\ &= \int e^{e^{ix}} (1 - e^{-ix}) dx \end{aligned}$$

Let $u = e^{ix}$ and $dx = \frac{du}{iu}$:

$$\begin{aligned} J &= \int e^u \left(1 - \frac{1}{u} \right) \frac{du}{iu} \\ &= -i \int e^u \left(\frac{1}{u} - \frac{1}{u^2} \right) du \end{aligned}$$

Notice that the integrand is the exact derivative of $\frac{e^u}{u}$ via the quotient (or product) rule:

$$\frac{d}{du} \left(\frac{e^u}{u} \right) = e^u \left(\frac{1}{u} - \frac{1}{u^2} \right)$$

Thus, the integral evaluates immediately:

$$\begin{aligned} J &= -i \left(\frac{e^u}{u} \right) \\ &= -ie^{e^{ix}} e^{-ix} \\ &= -ie^{\cos x + i \sin x - ix} \\ &= -ie^{\cos x} e^{i(\sin x - x)} \end{aligned}$$

To find I , we take the real part of J . Multiplying by $-i$ means we take the imaginary part of $e^{i(\sin x - x)}$.

$$\mathbf{I} = \mathbf{e}^{\cos \mathbf{x}} \sin(\sin \mathbf{x} - \mathbf{x}) + \mathbf{C}$$